

A Phase 2, Multicenter, Randomized, Controlled Clinical Trial Protocol

On-Premises LLM-Directed Robotic Pancreaticoduodenectomy (Whipple) with Perioperative Daraxonrasib (RMC-6236) in KRAS-Mutated Pancreatic Ductal Adenocarcinoma

Kevin Kawchak , CEO ChemicalQDevice | kevink@chemicalqdevice.com | 10.5281/zenodo.xxxxxxxx

Stage: Draft scaffold, v1.1.0 **Date:** June 23, 2026 **Pipeline:** mermaid → draft → full → final
Regulatory basis: IND (21 CFR part 312) + IDE (21 CFR part 812), Physical AI Subpart J **Predicate:** Phase 1 protocol (v1.0.0)

Draft scaffold (Stage 2). This file is not full prose: every NIH section below is a skeleton of bracketed [DRAFTING INSTRUCTION] blocks that name the exact repository source files, the Phase 1 model file, and the figures and tables the full stage must render. Independent research draft. Not an active IND or IDE, not an approved protocol, and not medical or regulatory advice; not endorsed by the FDA, NIH, HHS, an IRB, ICH, or any sponsor. The DOI placeholder 10.5281/zenodo.xxxxxxxx is filled at deposit. Clinical figures derive from the author's simulation sources and the Phase 1 predicate, and are illustrative unless tied to a cited reference.

Keywords Pancreatic Ductal Adenocarcinoma; Robotic Whipple; Daraxonrasib; Physical AI; On-premises LLM; Randomized Controlled Trial; Multicenter; IND/IDE; Phase 2 Protocol; Co-Investment; VVUQ

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1 Statement of Compliance

[DRAFTING INSTRUCTION] Follow the Phase 2 publication section `final-protocol/publication/sections/sec-00-compliance` and the Phase 1 model file `trial-protocol/draft-protocol/sections/sec-00-compliance.tex`. Render the dual-spine compliance statement for the randomized, controlled, multicenter efficacy study, framed as the trial the Phase 1 predicate deferred [1]. Source the section order and required content from `trial-protocol/nih-protocol/01_statement_of_compliance_protocol_summary_introduction.md`.

1.1 Drug Arm Compliance (IND, 21 CFR Part 312)

[DRAFTING INSTRUCTION] Carry the IND citations: 21 CFR parts 50, 54, 56, and 312, with the safety reporting of §312.32 and the amendment provisions of §312.20 through §312.31. State that, because the predicate established the RP2D, the drug arm is no longer a 3+3 evaluation; daraxonrasib proceeds at the fixed RP2D of 300 mg orally once daily as a Phase 2 controlled efficacy evaluation under §312.21(b). Source from `trial-protocol/inputs/21cfr312_adapt/02_ind_content_phases.tex` and `trial-protocol/inputs/2030-pdac-1min-final-paper/sections/introduction.tex`.

1.2 Device Arm Compliance (IDE, 21 CFR Part 812)

[DRAFTING INSTRUCTION] State the significant-risk IDE determination under §812.3(m) for the Pancre-Speed II eight-arm robotic Whipple, now framed as a multicenter randomized controlled clinical study under part 812; add the 21 CFR part 11 electronic-records clause supporting the federated hash-chained audit trail. Source from `trial-protocol/inputs/21cfr312_adapt/01_preamble_scope_definitions.tex`.

1.3 Physical AI Overlay (21 CFR Part 312, Subpart J)

[DRAFTING INSTRUCTION] Carry the Subpart J overlay (§312.400 through §312.405) with the upgraded Phase 0 gate: $USL \geq 8.0$, at least 5000 simulated procedures across at least 3 independent frameworks, $sim\text{-}to\text{-}real\text{ trajectory} < 1.5\text{ mm}$ and $tip\text{-}force < 0.4\text{ N}$, per-site qualification plus multicenter fleet harmonization, and a federated hash-chained audit trail under part 11. Retain Stage 2 supervised semiautonomy with $\leq 500\text{ ms e-stop}$; full autonomy prohibited under §312.21(e). Source from `trial-protocol/inputs/21cfr312_adapt/05_clinical_holds_appendices_closing.tex` and [2].

1.4 Good Clinical Practice and Electronic Records

[DRAFTING INSTRUCTION] Add the ICH E6(R3) GCP attestation and the 21 CFR part 11 electronic-records clause (validated systems, secure time-stamped audit trails, role-based access, bound electronic signatures) supporting the federated hash-chained record across all eight sites [3].

1.5 Multicenter Single IRB and Co-Investment Governance

[DRAFTING INSTRUCTION] State the single IRB of record (sIRB) for all eight sites under 45 CFR 46.114 [4], with reliance agreements coordinated by the Coordinating Center; disclose co-investment under 21 CFR part 54 [5] and the PACIF capital firewall governed by the H.R. 9510 VVUQ financial-data standard [6].

1.6 Trial Registration

[DRAFTING INSTRUCTION] Register as Interventional, randomized parallel-assignment; drug component Phase 2, device component Phase: Not Applicable, Primary Purpose Treatment; open-label with blinded independent central review and a prespecified group-sequential interim. Source from `trial-protocol/nih-protocol/01_statement_of_compliance_protocol_summary_introduction.md`.

1.7 Statement of Compliance Pathway

[DRAFTING INSTRUCTION] Render `trial-phase-2/mermaid/fig-01-ind-ide-pathway.md` here as a TikZ `mermaidfig` (this is **Figure 1**): the combined IND/IDE compliance spine with the upgraded Physical AI Subpart J overlay converging through the Phase 0 gate, USL readiness, single IRB across eight sites, and Class II collaborative classification into a single multicenter randomized enrollment pathway.

1.8 Signatures

[DRAFTING INSTRUCTION] Carry the signature blocks for the Sponsor-Investigator (Kevin Kawchak, CEO ChemicalQDevice), the Coordinating Principal Investigator, and the Multi-Site Principal Investigator (one line per participating site, eight HPB centers), with the independent-research-draft acknowledgment. Use `tabularx` with `L{}` and `Y` columns per the Phase 1 model.

2 Protocol Summary

2.1 Synopsis

[DRAFTING INSTRUCTION] Follow the Phase 2 publication section `final-protocol/publication/sections/sec-01-summary` and the Phase 1 model file `trial-protocol/draft-protocol/sections/sec-01-summary.tex`. Fill the Synopsis as a compressed mirror of the whole protocol: title, rationale, objectives and endpoints (from `sec-03`), design (multicenter, randomized 1:1, parallel-group, open-label with BICR, $n = 220$ at 110 per arm; from `sec-04` and `sec-09`), population (from `sec-05`), intervention (Arm A daraxonrasib at RP2D plus LLM-directed robotic Whipple; Arm B modified FOLFIRINOX plus standard Whipple; from `sec-06`), co-investment, and duration. Pull headline numbers from `trial-protocol/inputs/2030-pdac-1min-final-paper/sections/{introduction,results}.tex` and the Phase 1 predicate [1].

2.2 Schema

[DRAFTING INSTRUCTION] Render `trial-phase-2/mermaid/fig-02-trial-schema.md` as a TikZ `mermaidfig` (this is **Figure 2**): screening (28 days) -> eligibility -> central 1:1 stratified randomization with Physical AI opt-out -> two parallel arms -> follow-up with BICR -> PFS and 24-month OS endpoints, including the screen-failure branch.

2.3 Schedule of Activities (SoA)

[DRAFTING INSTRUCTION] Build the SoA as a full-width `tabularx` table at `\textwidth` with `L{}/C{}/Y` columns (**Table ??**). Columns: Screening (Day -28 to -1), Randomization / Baseline, Surgery (Day 0), Acute (Day 1 to 7), Day 30, Day 90, Long-term (q8-12wk). Rows: informed consent with Physical AI opt-out, staging / KRAS G12, CA 19-9, ECOG, safety labs, ctDNA (KRAS), central randomization, Phase 0 sign-off with $USL \geq 8.0$ (Arm A), surgery, intra-op telemetry (Arm A), ISGPS fistula grading, central R0 / MPR pathology, daraxonrasib restart advisory (Arm A), QoL EORTC QLQ-C30, BICR RECIST imaging, AE / SAE review, PFS / OS. Source the visit map from `trial-protocol/nih-protocol/01_statement_of_compliance_protocol_summary_introduction.md` and `trial-protocol/nih-protocol/06_study_assessments_and_procedures.md`.

3 Introduction

3.1 Study Rationale

[DRAFTING INSTRUCTION] Follow the Phase 2 publication section `final-protocol/publication/sections/sec-02-introduction` and the Phase 1 model file `trial-protocol/draft-protocol/sections/sec-02-introduction.tex`. State the PDAC burden (five-year OS < 13 percent [7]), the Whipple fistula-sepsis cascade [8], and the 2025 Dutch nationwide robotic cohort baseline [9]; establish equipoise from the Phase 1 predicate [1] and frame the randomized 1:1 comparison of Arm A versus Arm B (modified FOLFIRINOX [10]). Source from `trial-protocol/inputs/2030-pdac-1min-final-paper/sections/{introduction,discussion}.tex`.

[DRAFTING INSTRUCTION] Render `trial-phase-2/mermaid/fig-03-counterfactual-scenarios.md` as a TikZ `mermaidfig` (this is **Figure 3**): the four counterfactual scenarios (A resection-window collapse; B intra-op SMV/PV injury averted by the no-fly gate; C daraxonrasib restart mistiming; D under-funded under-powered study versus co-investment behind the firewall) in which withholding the integrated approach worsens the patient or the evidence.

3.2 Background

[DRAFTING INSTRUCTION] Carry the four Background subsubsections: PDAC epidemiology and the Whipple fistula-sepsis cascade; daraxonrasib mechanism and clinical program (FDA Break-through Therapy [11], RASolute 302 [12], and the five author simulation works [13, 14, 15, 16, 17]); current state of AI in clinical trials and surgery (FDA AI-device taxonomy [18] and the author LLM-trust record [19, 20, 2, 6]); and funding / co-investment success likelihood. Source from `trial-protocol/inputs/2030-pdac-1min-final-paper/sections/{introduction,methods,discussion}.tex` and `trial-protocol/inputs/author_works.bib`.

[DRAFTING INSTRUCTION] Render `trial-phase-2/mermaid/fig-04-physical-ai-concerns.md` as a TikZ mermaidfig (this is **Figure 4**): the nine Physical AI concerns mapped to their mitigations (the original eight device concerns plus the ninth, financial influence and inequitable access). State each pairing in full in the full-width **Table ??**. Source the concerns from `trial-protocol/inputs/21cfr312_adapt/01_preamble_scope_definitions.tex`.

[DRAFTING INSTRUCTION] Build the patient-aligned co-investment table (**Table ??**) mapping each tranche (eight-site network, Phase 0 compute, robot fleet plus redundancy, Patient Access and Equity Fund, central review and biomarker core, independent oversight) to the operational lever it funds and to the capital firewall, governed by H.R. 9510 and disclosed under 21 CFR part 54 [6, 5]. Source the firewall logic from `trial-protocol/inputs/auto-bill-02/final-bill/main.tex`.

3.3 Risk/Benefit Assessment

3.3.1 Known Potential Risks

[DRAFTING INSTRUCTION] Carry the three risk categories: drug risks (daraxonrasib toxicity at the RP2D, perioperative healing risk); device and robotic risks (malfunction, erroneous AI advisory, cyber-physical failure, vascular injury); and operative risks intrinsic to the Whipple (ISGPS fistula [8], hemorrhage, delayed gastric emptying, sepsis, conversion). Note that Arm B carries the operative and modified-FOLFIRINOX risks of standard care [10, 9].

3.3.2 Known Potential Benefits

[DRAFTING INSTRUCTION] Carry the directed benefits: in-window R0 resection; layered hard safety limits with a full hash-chained audit trail; optimized daraxonrasib restart timing ($T+7/T+14/T+21$); and the Patient Access and Equity Fund participant benefit.

3.3.3 Assessment of Potential Risks and Benefits

[DRAFTING INSTRUCTION] Render `trial-phase-2/mermaid/fig-05-risk-benefit.md` as a TikZ mermaidfig (this is **Figure 5**): weigh the novel risks against the benefits for this low-survival population and conclude a favorable balance consistent with 21 CFR §312.84. Then render `trial-phase-2/mermaid/fig-06-coinvestment-firewall.md` as a TikZ mermaidfig (this is **Figure 6**): the co-investment-to-success-likelihood pathway with the dashed blocked edge showing no funder path to endpoints, adjudication, or analysis [6, 5].

4 Objectives and Endpoints

[DRAFTING INSTRUCTION] Follow the Phase 2 publication section `final-protocol/publication/sections/sec-03-objectives` and the Phase 1 model file `trial-protocol/draft-protocol/sections/sec-03-objectives.tex`. Build the governing three-column objectives / endpoints / justification table (**Table ??**) or-organized as one confirmatory efficacy question, a fixed-sequence key secondary hierarchy, estimation-only secondary measures, and exploratory measures. Source the structure from

trial-protocol/nih-protocol/02_objectives_and_endpoints.md. Carry the abbreviations line and note the internal control (Arm B) versus the external Dutch descriptive benchmark [9].

[DRAFTING INSTRUCTION] Render *trial-phase-2/mermaid/fig-07-objectives-endpoints.md* as a TikZ *mermaidfig* (this is **Figure 7**): the objective-to-endpoint hierarchy with PRIMARY (PFS, Arm A vs Arm B) above the fixed-sequence KEY SECONDARY family (OS; R0; ISGPS B/C; MPR; ctDNA) above the estimation-only SECONDARY and the EXPLORATORY measures, with dashed tier separators.

4.1 Primary Objective and Endpoint

[DRAFTING INSTRUCTION] Single primary objective: compare PFS between Arm A and Arm B. Primary endpoint PFS per RECIST 1.1, investigator-assessed with BICR as a prespecified sensitivity analysis; sized for ~ 140 events for 85 percent power at two-sided $\alpha = 0.05$ to detect HR 0.60 (median PFS 14.0 vs 23.3 months), with one group-sequential interim [21, 22] (§9).

4.2 Key Secondary Objectives and Endpoints

[DRAFTING INSTRUCTION] Fixed-sequence gatekeeping hierarchy controlling the family-wise type I error at two-sided 0.05: (1) OS; (2) R0 resection rate; (3) ISGPS grade B/C fistula [8]; (4) MPR; (5) week-12 KRAS ctDNA clearance [23]. First non-significant result closes the sequence.

4.3 Secondary Objectives and Endpoints

[DRAFTING INSTRUCTION] Estimation-only measures with two-sided 95 percent CIs: Clavien-Dindo grade III+ through 90 days (Dutch comparator 41.3 percent), 90-day mortality, unplanned conversion rate (Dutch comparator 10.1 percent), operative time / EBL / LOS, and quality of life (EORTC QLQ-C30 and PAN26 [24]) with 12-, 18-month PFS and 24-month OS rates [9].

4.4 Exploratory Objectives and Endpoints

[DRAFTING INSTRUCTION] Non-confirmatory measures: LLM advisory-to-gate concordance; sim-to-real gap (trajectory mm, tip-force N) against the upgraded Phase 0 thresholds; federated-learning model performance and ctDNA dynamics across the eight sites [23]; and health-economic / cost-effectiveness outcomes linked to the co-investment design.

5 Study Design

5.1 Overall Design

[DRAFTING INSTRUCTION] Follow the Phase 2 publication section *final-protocol/publication/sections/sec-04-design* and the Phase 1 model file *trial-protocol/draft-protocol/sections/sec-04-design.tex*. Describe the prospective, multicenter, randomized, parallel-group, controlled, open-label design with BICR across eight academic HPB centers; $n = 220$ randomized 1:1 (110 per arm); Arm A daraxonrasib RP2D 300 mg PO daily plus PancreSpeed II, Arm B modified FOLFIRINOX [10] plus standard Whipple; central permuted-block randomization stratified by resectability, KRAS allele, neoadjuvant therapy, and site. Source the design and population order from *trial-protocol/nih-protocol/03_study_design_and_study_population.md*.

[DRAFTING INSTRUCTION] Render *trial-phase-2/mermaid/fig-08-randomization-design.md* as a TikZ *mermaidfig* (this is **Figure 8**): eligible participant $\rightarrow$ central stratified randomization 1:1 $\rightarrow$ Arm A / Arm B $\rightarrow$ eight harmonized HPB sites $\rightarrow$ BICR plus central masked pathology $\rightarrow$ endpoint adjudication.

5.2 Scientific Rationale for Study Design

[DRAFTING INSTRUCTION] Justify the randomized controlled design as the ethical next step that inverts the single-arm predicate [1]; open-label conduct is unavoidable for a surgical-system intervention, so bias control is concentrated downstream (BICR imaging, central masked R0/MPR pathology, blinded adjudication). The internal control Arm B replaces the external Dutch benchmark [9, 7]. Follow the CONSORT framework [25].

5.3 Justification for Dose and Device Readiness

[DRAFTING INSTRUCTION] *Daraxonrasib at the established RP2D of 300 mg once daily (no 3+3 escalation), anchored to RASolute 302 [12, 1]. The device “dose” analogue is the upgraded Phase 0 gate: $USL \geq 8.0$, at least 5000 sims across at least 3 frameworks, sim-to-real < 1.5 mm and < 0.4 N, per-site qualification plus fleet harmonization, operating with the intra-op hard caps (≤ 3 N per-arm, ≤ 18 N cumulative, ≤ 3 ms e-stop, five-vessel no-fly gate) [26, 2]. Source from `trial-protocol/inputs/21cfr312_adapt/02_ind_content_phases.tex`.*

[DRAFTING INSTRUCTION] *Render `trial-phase-2/mermaid/fig-09-staged-autonomy.md` as a TikZ `mermaidfig` (this is **Figure 9**): the phase-graduated staged-autonomy model running at Stage 2 supervised semiautonomy (Class II, continuous oversight, immediate override); any expanded subtask requires independent safety review and $USL \geq 8.0$; Stage 3 full autonomy reserved and prohibited under 21 CFR §312.21(e).*

5.4 End of Study Definition

[DRAFTING INSTRUCTION] *Define individual completion at the 24-month OS assessment (or death, progression precluding follow-up, or withdrawal), and overall end of study as the last scheduled PFS/OS assessment plus primary-analysis database lock after ~ 140 PFS events. Discontinued participants (§7) remain in long-term survival follow-up.*

6 Study Population

[DRAFTING INSTRUCTION] *Follow the Phase 2 publication section `final-protocol/publication/sections/sec-05-pop` and the Phase 1 model file `trial-protocol/draft-protocol/sections/sec-05-population.tex`. Describe the population: adults with histologically confirmed, resectable or borderline-resectable, KRAS G12-mutated PDAC, daraxonrasib-eligible under broad pan-KRAS criteria [12], across eight HPB centers; ~ 245 screened to yield 220 randomized 1:1 (110 per arm) allowing up to 10 percent non-evaluability (§9). Source from `trial-protocol/nih-protocol/03_study_design_and_study_population.md`.*

[DRAFTING INSTRUCTION] *Render `trial-phase-2/mermaid/fig-10-consort-flow.md` as a TikZ `mermaidfig` (this is **Figure 10**): the randomized two-arm CONSORT participant flow from ~ 245 assessed through exclusions, 220 randomized 1:1, Arm A / Arm B (~ 110 each), the discontinuation branch (§7), and the ITT, mITT, per-protocol, and safety analysis populations [25].*

6.1 Inclusion Criteria

[DRAFTING INSTRUCTION] *Enumerate the inclusion criteria: signed consent with the Physical AI opt-out election; age ≥ 18 ; confirmed resectable / borderline-resectable PDAC; documented KRAS G12 mutation; ECOG 0 or 1; adequate organ and marrow function; measured baseline CA 19-9; daraxonrasib-eligible per RASolute 302 [12]; and anatomy compatible with the eight-arm robotic system for those who may receive Arm A. Include the synthetic exemplar PAT-PDAC-0002 (66-year-old, 3.2 cm head-of-pancreas tumor abutting the SMV, KRAS G12D) after neoadjuvant modified FOLFIRINOX [10].*

6.2 Exclusion Criteria

[DRAFTING INSTRUCTION] *Enumerate the exclusion criteria: distant metastatic disease; vascular involvement precluding R0; prior pancreatic resection precluding reconstruction; uncontrolled intercurrent illness; pregnancy or lactation; contraindication or hypersensitivity to daraxonrasib or modified FOLFIRINOX; and inability to consent or comply. State equitable selection and the Patient Access and Equity Fund behind the capital fire-wall [6, 5].*

6.3 Lifestyle Considerations

[DRAFTING INSTRUCTION] *Perioperative nutrition plan, alcohol avoidance during dosing, tobacco cessation support, routine-activity guidance, and the daraxonrasib labeling interactions deferred to the concomitant-medication rules in §6.*

6.4 Screen Failures

[DRAFTING INSTRUCTION] Define screen failure and the minimal CONSORT screen-failure data set; allow rescreening of transient disqualifiers with the same participant number, so the screening-to-randomization accounting in Figure 10 stays complete.

6.5 Strategies for Recruitment and Retention

[DRAFTING INSTRUCTION] Recruitment across eight HPB centers at ~ 10 to 14 participants per month over ~ 30 months; consent presented by a non-treating team member; retention through structured follow-up and the Patient Access and Equity Fund behind the capital firewall (travel, lodging, lost-wage, caregiver, navigation), with OHRP-consistent safeguards and no undue inducement [6, 5].

7 Study Intervention

[DRAFTING INSTRUCTION] Follow the Phase 2 publication section `final-protocol/publication/sections/sec-06-int` and the Phase 1 model file `trial-protocol/draft-protocol/sections/sec-06-intervention.tex`. Describe both randomized arms under one integrated protocol; Arm A couples the drug and device at the single perioperative pause-and-restart decision point (§6.1.2). Source the intervention order from `trial-protocol/nih-protocol/04_study_intervention.md`.

7.1 Study Intervention Administration

7.1.1 Study Intervention Description

[DRAFTING INSTRUCTION] Describe the three components. Arm A drug: daraxonrasib (RMC-6236), RAS(ON) multi-selective inhibitor, oral, perioperative [11, 12]. Arm B control: modified FOLFIRINOX [10] plus institutional-standard high-volume pancreaticoduodenectomy. Arm A device: the eight-arm LLM-directed robotic Whipple platform (PancreSpeed II) - 56 DOF, 640 sensor channels, 0.05 mm RMS at 1200 mm/s, ≤ 3 N per-arm and ≤ 18 N cumulative force caps, 10 kHz heartbeat with 100 μ s watchdog, ≤ 3 ms cross-arm e-stop, five-vessel no-fly gate, fleet-harmonized, Stage 2 supervised semiautonomy. Source from `trial-protocol/inputs/2030-pdac-1min-final-paper/sections/methods.tex` and the 21 CFR 312.23(g) eleven-field Physical AI System Description from `trial-protocol/inputs/21cfr312_adapt/02_ind_content_phases`.

[DRAFTING INSTRUCTION] Render `trial-phase-2/mermaid/fig-11-llm-advisory-loop.md` as a TikZ mermaidfig (this is **Figure 11**): the on-premises LLM advisory control loop (sensor $\rightarrow$ Cartesian frame $\rightarrow$ hash-pinned LLM advisory outside the vendor stack $\rightarrow$ safety gate $\rightarrow$ actuators $\rightarrow$ hash-chained federated audit under 21 CFR part 11 [3]). Then render `trial-phase-2/mermaid/fig-12-platform-architecture.md` as a TikZ mermaidfig (this is **Figure 12**): the eight-arm platform architecture coordinated by the 10 kHz heartbeat bus.

[DRAFTING INSTRUCTION] Build two full-width tables: the per-arm tool assignment crossed with the eight operative phases P1-P8 (**Table ??**) and ten representative channels from the 640-channel inventory (**Table ??**). Source both from `trial-protocol/inputs/2030-pdac-1min-final-paper/sections/methods.tex`.

7.1.2 Dosing and Administration

[DRAFTING INSTRUCTION] Daraxonrasib at the fixed RP2D of 300 mg PO once daily (no 3+3 escalation), anchored to RASolute 302 [1, 12]. Render `trial-phase-2/mermaid/fig-13-daraxonrasib-advisory.md` as a TikZ mermaidfig (this is **Figure 13**): the perioperative pause-and-restart advisory keyed to the PJ ISGPS fistula grade [8] and the 0.5 ng/mL serum trough (restart T+7 / T+14 / T+21 or continued hold). Then render `trial-phase-2/mermaid/fig-14-three-anastomoses.md` as a TikZ mermaidfig (this is **Figure 14**): the device administration across the eight operative phases and the three reconstructive anastomoses (PJ 0.30 to 0.60 N, HJ 0.20 to 0.50 N, GJ 0.40 to 0.80 N) under Arm 5 ring-tension control.

7.2 Preparation, Handling, Storage, and Accountability

7.2.1 Acquisition and Accountability

[DRAFTING INSTRUCTION] Sponsor-supplied daraxonrasib dispensed against a written prescription tied to an enrolled Arm A participant with a drug-accountability log; Arm B modified FOLFIRINOX via standard oncology pharmacy; the device accounted for via the asset record, pre-procedure safety matrix, and federated audit trail (§6.4).

7.2.2 Formulation, Appearance, Packaging, and Labeling

[DRAFTING INSTRUCTION] Daraxonrasib oral tablets composing the 300 mg RP2D, labeled per 21 CFR §312.6 with the investigational caution statement; the device and single-use accessories labeled per 21 CFR part 812 and not relabeled at site.

7.2.3 Product Storage and Stability

[DRAFTING INSTRUCTION] Controlled-room-temperature, access-controlled, temperature-monitored storage with excursion handling; device stored and maintained in its qualified clinical environment under the maintenance program, identically across the fleet.

7.2.4 Preparation and Device Maintenance and Calibration

[DRAFTING INSTRUCTION] No drug compounding; tablets swallowed whole. Device maintenance and calibration per 21 CFR §312.23(g)(xi): the pre-procedure safety matrix (640-channel verification, per-arm force calibration against the $\leq 3\text{ N} / \leq 18\text{ N}$ caps, 10 kHz heartbeat with 100 μs watchdog, five-vessel no-fly gate, software and hardware e-stop self-test, 0.05 mm RMS positioning), with per-site qualification and fleet harmonization, all logged to the audit trail.

7.3 Measures to Minimize Bias: Randomization and Blinding

[DRAFTING INSTRUCTION] Central permuted-block randomization 1:1 stratified by resectability, KRAS allele, neoadjuvant therapy, and site [25]; open-label delivery with downstream bias control (BICR imaging, central masked R0/MPR pathology with independent adjudication, blinded endpoint adjudication). Cite ICH E9 and `trial-protocol/nih-protocol/04_study_intervention.md`.

7.4 Study Intervention Compliance

[DRAFTING INSTRUCTION] Daraxonrasib adherence via the dispensing log, returned-tablet counts, diary, and serum assay (which supplies the advisory trough); Arm B via cycle administration records. Device compliance enforced before and during every procedure and written to the 21 CFR part 11 hash-chained federated audit trail [3]. (The audit-trail data flow is rendered as Figure 21 in §10.)

7.5 Concomitant Therapy

[DRAFTING INSTRUCTION] Permitted perioperative supportive care in both arms with attention to daraxonrasib interactions in Arm A; prohibited systemic anticancer therapy outside the assigned regimen, other investigational agents or devices, and a second surgical robot for the index operation.

7.5.1 Rescue Medicine

[DRAFTING INSTRUCTION] *Rescue along two axes: surgical / device complications (conversion to manual or open per §7, transfusion, drainage, antimicrobials, somatostatin analogue) and drug toxicity (hold or discontinue daraxonrasib per §7; dose-modify modified FOLFIRINOX per its established management). Record all rescue as concomitant therapy and, where applicable, as AE / SAE.*

8 Intervention and Participant Discontinuation/Withdrawal

[DRAFTING INSTRUCTION] *Follow the Phase 2 publication section `final-protocol/publication/sections/sec-07-discontinuation` and the Phase 1 model file `trial-protocol/draft-protocol/sections/sec-07-discontinuation.tex`. No figure or table is rendered in this section. Distinguish discontinuation of an intervention from withdrawal from the study: a participant who stops one or both components for a protocol-defined reason remains in the randomized ITT population for survival follow-up. Apply rules within the randomized arm. Source from `trial-protocol/nih-protocol/05_intervention_and_participant_discontinuation.md`.*

8.1 Discontinuation of Study Intervention

[DRAFTING INSTRUCTION] *Device stopping rules (Arm A), implementing 21 CFR §312.23(g)(i)(v) and the divergence-reporting threshold of §312.32(g) [2]: conversion to open or manual technique; repeated force-limit excursions against the $\leq 3\text{ N} / \leq 18\text{ N caps}$; no-fly-zone violations (SMV/PV 1.0 mm; HA/CA/SMA 1.5 mm); sim-to-real divergence beyond twice tolerance ($> 3\text{ mm}$ or $> 0.8\text{ N}$); and any hard-stop e-stop. Drug stopping rules: daraxonrasib discontinued for unresolved toxicity, confirmed ISGPS Grade C fistula [8], pregnancy, contraindicating illness, or request; modified FOLFIRINOX managed per standard toxicity rules [10]. The federated hash-chained audit trail is preserved across each event (§10) and reviewed by the Physical AI Safety Review Committee and the DSMB before further device use.*

8.2 Participant Discontinuation/Withdrawal from the Study

[DRAFTING INSTRUCTION] *Voluntary withdrawal at any time without penalty; separate withdrawal of consent for procedures while permitting survival follow-up; investigator-initiated withdrawal for safety or non-compliance. Data and specimens collected up to withdrawal are retained. Because this is a randomized ITT trial, participants are analyzed as randomized and are not replaced (§9).*

8.3 Lost to Follow-Up

[DRAFTING INSTRUCTION] *Define lost-to-follow-up only after at least three documented telephone attempts across separate days plus a written attempt over the 24-month OS window, with vital-status ascertainment from the record, referring provider, and permissible public sources; record the last date known alive. Missing-data and censoring handling is specified in §9.*

9 Study Assessments and Procedures

[DRAFTING INSTRUCTION] *Follow the Phase 2 publication section `final-protocol/publication/sections/sec-08-assessments` and the Phase 1 model file `trial-protocol/draft-protocol/sections/sec-08-assessments.tex`. State that every centrally readable endpoint is read centrally across the eight sites (central imaging core BICR, central masked pathology, central laboratories). Timing is governed by the SoA (§1.3). Source from `trial-protocol/nih-protocol/06_study_assessments_and_procedures.md`.*

9.1 Efficacy Assessments

[DRAFTING INSTRUCTION] *Define PFS (RECIST 1.1, investigator-assessed primary with BICR sensitivity, §9.4.2) and OS to 24 months [9]; the central masked R0 resection rate and MPR; ISGPS biochemical-leak / Grade B / Grade C fistula grading [8]; week-12 KRAS ctDNA clearance [23]; the Arm A anastomosis-quality reads against the ring-tension bands; and QoL via EORTC QLQ-C30 and PAN26 [24].*

9.2 Safety and Other Assessments

[DRAFTING INSTRUCTION] *Clinical safety (vitals, exam, hematology, chemistry, coagulation, the Arm A serum daraxonrasib trough, imaging) plus the Arm A continuous device-telemetry stream under 21 CFR §312.23(g) (per-arm and cumulative force profiles, trajectory accuracy, vascular-zone verdicts, heartbeat / watchdog events, e-stop activations). Render `trial-phase-2/mermaid/fig-15-vascular-safety-gate.md` as a TikZ mermaidfig (this is **Figure 15**): the five-vessel vascular safety-zone gate (soft-warning 3.0 mm, no-fly 1.0/1.5 mm, hard-stop 5.0 mm) sampled at 10 kHz. Then render `trial-phase-2/mermaid/fig-16-estop-architecture.md` as a TikZ mermaidfig (this is **Figure 16**): the heartbeat / watchdog / e-stop architecture (100 μ s watchdog, 50 μ s arm park, ≤ 3 ms cross-arm e-stop, ≤ 500 ms hardware e-stop). Source from `trial-protocol/inputs/2030-pdac-1min-final-paper/sections/methods.tex` and `trial-protocol/inputs/21cfr312_adapt/05_clinical_holds_appendices_closing.tex`.*

9.3 Adverse Events and Serious Adverse Events

[DRAFTING INSTRUCTION] *Two parallel binding streams: conventional pharmacovigilance for both arms and a Physical AI safety stream for Arm A. Render `trial-phase-2/mermaid/fig-17-ae-reporting.md` as a TikZ mermaidfig (this is **Figure 17**): the combined AE / Physical AI AE reporting workflow (CTCAE v5 grading, 7-day fatal / 15-day serious-unexpected timelines, the six Physical AI triggers, audit-trail preservation –24 h to +72 h federated across sites, and notification of FDA / IRB / DSMB / Safety Review Committee). Source from `trial-protocol/inputs/21cfr312_adapt/03_protocol_amendments_reporting.tex`.*

9.3.1 Definition of Adverse Events (AE)

[DRAFTING INSTRUCTION] *Standard AE definition, CTCAE v5 grading in both arms, plus the Physical AI adverse event per 21 CFR §312.32(f) (robotic malfunction, AI prediction error, sensor failure, communication loss, unauthorized access, digital-twin divergence, any e-stop activation) [2].*

9.3.2 Definition of Serious Adverse Events (SAE)

[DRAFTING INSTRUCTION] *Standard SAE criteria plus the serious Physical AI adverse event per §312.32(f) (potential SAE outcome, unplanned interruption of drug administration, uncontrolled drug release, or sterile-field compromise).*

9.3.3 Classification of an Adverse Event

[DRAFTING INSTRUCTION] *Four axes: severity (CTCAE v5); separate drug and device causality (procedure causality in Arm B); expectedness against the reference safety information; and seriousness. Unexpected + serious + at least possibly related defines a reportable suspected adverse reaction.*

9.3.4 Time Period and Frequency for Assessment and Follow-Up

[DRAFTING INSTRUCTION] *Solicit AEs from consent; intensive window through the 30-day visit; SAEs, events of special interest, Physical AI AEs, and new anticancer therapy collected to the 24-month OS endpoint.*

9.3.5 Adverse Event Reporting

[DRAFTING INSTRUCTION] *Record all AEs in the CRF with onset / resolution, grade, drug and device causality, expectedness, seriousness, action, and outcome; Arm A telemetry- derived observations cross-referenced to audit-trail records.*

9.3.6 Serious Adverse Event Reporting

[DRAFTING INSTRUCTION] Investigator-to-sponsor within 24 hours; FDA timelines (7-day fatal / life-threatening, 15-day other serious unexpected) plus the six Physical AI reporting triggers under §312.32(g); preserve the hash-chained audit trail -24 h to $+72$ h, federated across the eight sites [3].

9.3.7 Events of Special Interest

[DRAFTING INSTRUCTION] Vascular injury (including any vascular-breach hard-stop e-stop), clinically relevant fistula (ISGPS B/C [8]), conversion, any cybersecurity incident, and any model-performance degradation; reviewed by the Physical AI Safety Review Committee (≤ 90 -day cadence) and reported to the DSMB.

9.3.8 Reporting of Pregnancy

[DRAFTING INSTRUCTION] Pregnancy reported (not an AE) within 24 hours and followed to outcome; highly effective contraception required; an associated SAE reported on the expedited timeline.

9.4 Unanticipated Problems

9.4.1 Definition

[DRAFTING INSTRUCTION] Standard unanticipated-problem definition; treat the Physical AI hold grounds (repeated uncorrected safety failures, confirmed cybersecurity compromise, simulation-validation deficiency) as paradigm unanticipated problems.

9.4.2 Reporting

[DRAFTING INSTRUCTION] Report to the single IRB and, as applicable, the sponsor and FDA on the 7- or 15-day SAE timeline or promptly otherwise; device-related reports preserve the federated audit trail [3]; the Physical AI Safety Review Committee and DSMB review each.

9.4.3 Reporting to Participants

[DRAFTING INSTRUCTION] Communicate to affected participants via IRB-approved re-consent or written notification; update the consent and re-consent enrolled participants when the risk profile changes, consistent with the Physical AI opt-out.

10 Statistical Considerations

[DRAFTING INSTRUCTION] Follow the Phase 2 publication section `final-protocol/publication/sections/sec-09-statistics.tex` and the Phase 1 model file `trial-protocol/draft-protocol/sections/sec-09-statistics.tex`. Frame this as a confirmatory, randomized, controlled superiority trial with type I error protection (contrasting the descriptive single-arm predicate [1]); follow ICH E9 (including the estimand framework), CONSORT [25], and TRIPOD+AI [27]. Source from `trial-protocol/nih-protocol/07_statistical_considerations.md`.

10.1 Statistical Hypotheses

[DRAFTING INSTRUCTION] Primary PFS hypothesis $H_0 : \theta \geq 1$ vs $H_1 : \theta < 1$, stratified log-rank with stratified Cox HR and two-sided 95 percent CI, two-sided $\alpha = 0.05$ allocated across one interim and the final analysis by alpha spending. Key secondary fixed-sequence hierarchy (OS; R0; ISGPS B/C [8]; MPR; week-12 KRAS ctDNA [23]) controlling the family-wise error at two-sided 0.05.

10.2 Sample Size Determination

[DRAFTING INSTRUCTION] $n = 220$ randomized 1:1 (110 per arm), event-driven: ~ 140 PFS events give 85 percent power at two-sided $\alpha = 0.05$ for HR 0.60 (median PFS 14.0 vs 23.3 months); up to ~ 245 enrolled to allow ≤ 10 percent non-evaluability; one interim at ~ 60 percent of events (~ 84) with a Lan-DeMets O'Brien-Fleming spending function and non-binding futility boundary [21, 22]. Summarize the operating characteristics in the full-width **Table ??**.

10.3 Populations for Analyses

[DRAFTING INSTRUCTION] Define ITT (all randomized, primary), modified ITT (dosed with ≥ 1 post-baseline assessment), per-protocol, and safety (intervention as received) populations. Render `trial-phase-2/mermaid/fig-18-analysis-populations.md` as a TikZ `mermaidfig` (this is **Figure 18**): the four populations and the single group-sequential interim with the O'Brien-Fleming efficacy / futility boundary and the parallel binding device-safety halt rules.

10.4 Statistical Analyses

10.4.1 General Approach

[DRAFTING INSTRUCTION] Confirmatory primary and key secondary analyses with type I error protection; estimation-based descriptive remainder. Stratified log-rank with Kaplan-Meier and stratified Cox HR for time-to-event (RMST fallback if proportional hazards fails); stratified Cochran-Mantel-Haenszel for confirmatory binary endpoints; ICH E9 estimands for intercurrent events; deterministic-seed, version-controlled, reproducible analysis.

10.4.2 Analysis of the Primary Endpoint

[DRAFTING INSTRUCTION] PFS on ITT by stratified log-rank at the alpha allocated to the final analysis, with the stratified Cox HR and 95 percent CI; BICR-assessed PFS as the prespecified sensitivity analysis, plus per-protocol, mITT, and alternative-censoring sensitivity analyses (§8.1).

10.4.3 Analysis of the Key Secondary Endpoints

[DRAFTING INSTRUCTION] Fixed-sequence hierarchy (OS, R0, ISGPS B/C, MPR, week-12 ctDNA), each at two-sided 0.05, stopping at the first non-rejection. Summarize the key secondary and other secondary endpoints, populations, methods, and the external Dutch 2025 descriptive context in the full-width **Table ??** [9, 23].

10.4.4 Safety Analyses

[DRAFTING INSTRUCTION] Safety population; MedDRA-coded AEs by system organ class and CTCAE severity with drug, device (Arm A), and procedure causality by arm; the parallel Arm A cyber-physical telemetry summaries (e-stop latency vs ≤ 500 ms, force excursions vs the $\leq 3 N / \leq 18 N$ caps, vascular-zone gating, advisory accept / block / escalate counts) and the Physical AI AE tabulation keyed to the federated audit trail (§8.3.6).

10.4.5 Baseline Descriptive Statistics

[DRAFTING INSTRUCTION] Baseline characteristics by arm and overall (demographics, staging, resectability, KRAS allele, CA 19-9, ECOG, neoadjuvant cycles) without a baseline significance test, per CONSORT [25].

10.4.6 Planned Interim Analyses

[DRAFTING INSTRUCTION] Single formal interim at ~ 60 percent of PFS events (~ 84), DSMB-reviewed against the Lan-DeMets O’Brien-Fleming efficacy and non-binding futility boundaries [21, 22], with parallel binding device-safety halt rules under §312.30(h).

10.4.7 Sub-Group Analyses

[DRAFTING INSTRUCTION] Prespecified subgroups (resectability, KRAS allele, neoadjuvant, site) with a forest plot and interaction test, interpreted as supportive and not adjusted for multiplicity.

10.4.8 Tabulation of Individual Participant Data

[DRAFTING INSTRUCTION] Participant-level listings (demographics, arm and strata, actual intervention, all AEs, SAEs and deaths, efficacy endpoints with dates, and the Arm A per-procedure device-telemetry and Physical AI event summaries).

10.4.9 Exploratory Analyses

[DRAFTING INSTRUCTION] Non-confirmatory: LLM advisory-to-gate concordance, sim-to-real gap, federated-learning model performance across sites, longitudinal ctDNA dynamics [23], and health-economic / cost-effectiveness linked to the co-investment facility.

11 Regulatory, Ethical, and Oversight Considerations

[DRAFTING INSTRUCTION] Follow the Phase 2 publication section `final-protocol/publication/sections/sec-10-over` and the Phase 1 model file `trial-protocol/draft-protocol/sections/sec-10-oversight.tex`. Open with the multicenter oversight architecture (Coordinating Center, Sponsor-Investigator, single IRB under 45 CFR 46.114 [4], DSMB with a group-sequential interim, Physical AI Safety Review Committee on a ≤ 90-day cadence, per-procedure Independent Safety Monitor, and a robotics- and AI-competent CRO). Render `trial-phase-2/mermaid/fig-19-governance-oversight.md` as a TikZ `mermaidfig` (this is **Figure 19**). Source the oversight order from `trial-protocol/nih-protocol/08_supporting_documentation_regulatory_oversight.tex` and duties from `trial-protocol/inputs/21cfr312_adapt/04_annual_reports_withdrawal.tex`.

11.1 Informed Consent Process

[DRAFTING INSTRUCTION] Describe consent by a non-treating team member under 21 CFR part 50 [28] and ICH E6(R3), a single master consent across the network, disclosure of the randomized open-label allocation and the LLM-directed system, and the documented Physical AI opt-out under 21 CFR §312.60(f) [2]. Render `trial-phase-2/mermaid/fig-20-informed-consent.md` as a TikZ `mermaidfig` (this is **Figure 20**). Source from `trial-protocol/inputs/21cfr312_adapt/05_clinical_holds_appendices_closing.tex`.

11.2 Funding, Co-Investment Governance, and the Capital Firewall

[DRAFTING INSTRUCTION] Describe the PACIF funding only the six operational levers and never scientific outcomes, the capital firewall walling funders off from randomization, endpoints, adjudication, analysis, data access, and publication, disclosure under 21 CFR part 54 [5], governance by the H.R. 9510 VVUQ financial-data standard [6], and open CC BY 4.0 deposition. Render `trial-phase-2/mermaid/fig-21-capital-firewall.md` as a TikZ `mermaidfig` (this is **Figure 21**): the capital-firewall governance with the dashed blocked path to the science. Source the firewall model from `trial-protocol/inputs/auto-bill-02/final-bill/main.tex`.

11.3 Study Discontinuation and Closure

[DRAFTING INSTRUCTION] *Closure by the Sponsor-Investigator, single IRB, or FDA; the Physical AI clinical-hold grounds (hard cyber-physical limit breach, verification-before-generation gate-surface failure, loss of audit integrity) that may halt one site or the fleet; and the controlled decommissioning and data-preservation procedure (§10.10).*

11.4 Confidentiality and Privacy

[DRAFTING INSTRUCTION] *HIPAA Safe Harbor de-identification (18 identifiers), deny-by-default authorization, and the federated hash-chained audit trail under 21 CFR part 11 [3]. (The audit-trail data flow is the Figure 21 capital-firewall companion logic; the hash-chained loop itself is rendered in Figure 11, §6.)*

11.5 Future Use of Stored Specimens and Data

[DRAFTING INSTRUCTION] *The tiered telemetry pyramid (apex summaries and safety / audit logs; middle compressed feature streams; L0 raw losslessly reconstructable), L0 deposited on Zenodo and decompressible on FDA request per 21 CFR §312.57; ctDNA and biomarker specimens governed by consent.*

11.6 Key Roles and Study Governance

[DRAFTING INSTRUCTION] *Sponsor-Investigator (ChemicalQDevice, Kevin Kawchak) holding the combined IND/IDE under parts 312 and 812; Coordinating Center; site PIs; the Arm A operating surgeon, primary operator, and backup operator under supervised semiautonomy; the CRO; and the DSMB / ISM / Physical AI Safety Review Committee, on a delegation-of-authority log.*

11.7 Safety Oversight

[DRAFTING INSTRUCTION] *Three tiers: the DSMB (group-sequential interim with O'Brien-Fleming spending [21] and the §9 stopping rules), the per-procedure Independent Safety Monitor, and the Physical AI Safety Review Committee (≤ 90 -day cadence) reviewing the fleet telemetry, gate-surface decisions, and USL rating.*

11.8 Clinical Monitoring

[DRAFTING INSTRUCTION] *Robotics- and AI-competent CRO monitors; risk-based plan; source-data verification of clinical documents, consent (including the Arm A Physical AI opt-out election), and the cyber-physical evidence (telemetry, gate-surface logs, federated audit trail).*

11.9 Quality Assurance and Quality Control

[DRAFTING INSTRUCTION] *Pre-procedure safety-matrix verification and USL confirmation before every Arm A case; Phase 0 re-validation, USL reassessment (≥ 8.0), and fleet re-harmonization after any software, model, or configuration change before further patient-facing use; deterministic version-controlled build.*

11.10 Data Handling and Record Keeping

[DRAFTING INSTRUCTION] *The seven Physical AI record types (LLM advisory; robot command; sensor / telemetry; safety-limit-event; verification-before-generation gate-surface; human-override; and the federated hash-chained audit trail binding the first six), retained per 21 CFR §312.57 with the L0 raw tier preserved (§10.5).*

11.11 Protocol Deviations

[DRAFTING INSTRUCTION] *Definition, documentation, assessment, and reporting to the single IRB and Sponsor-Investigator; immediate-hazard modifications under 21 CFR §312.30(b), which for the autonomy-bearing system also trigger the decommissioning and re-validation controls of §10.4 and §10.11.*

11.12 Publication and Data Sharing Policy

[DRAFTING INSTRUCTION] *Dual deposition (GitHub plus Zenodo, CC BY 4.0) regardless of outcome; reporting per TRIPOD+AI [27] and CONSORT [25]; published deterministic seed and repository commit; HIPAA Safe Harbor de-identification (§10.5).*

11.13 Conflict of Interest Policy

[DRAFTING INSTRUCTION] *Disclose and manage financial interests under 21 CFR part 54 [5]; the Sponsor-Investigator's ChemicalQDevice relationship managed by the independent oversight of §10.9 and the capital firewall of §10.3; certifications collected before participation and updated through one year after the study.*

12 Additional Considerations, Abbreviations, and Amendment History

12.1 Additional Considerations

[DRAFTING INSTRUCTION] *Follow the Phase 2 publication section `final-protocol/publication/sections/sec-11-add` and the Phase 1 model file `trial-protocol/draft-protocol/sections/sec-11-additional.tex`. Describe the cross-jurisdictional SaMD and robotically assisted surgery tracks (EU MDR plus AI requirements, NMPA, PMDA, MHRA, Health Canada, TGA) onto which the common technical spine maps without redesign, anchored to the consensus standards [29, 30, 31, 32], and summarize them in the full-width **Table ??**. Describe the federated multicenter learning model and the H.R. 9510 VVUQ alignment [6]. Source from `trial-protocol/nih-protocol/09_additional_considerations_abbreviations_amendment_history.md` and the ten-gate logic from `trial-protocol/inputs/auto-bill-02/final-bill/main.tex`.*

[DRAFTING INSTRUCTION] *Render `trial-phase-2/mermaid/fig-22-vvuq-ten-gate.md` as a TikZ `mermaidfig` (this is **Figure 22**): verification before generation - every proposed robot-patient command passes a ten-gate assurance suite (fourteen external standards plus two clinical baselines) with epistemic and aleatory uncertainty bounds before the gate surface returns ACCEPT, BLOCK, or ESCALATE.*

12.2 Protocol Amendment History

[DRAFTING INSTRUCTION] *Build the amendment-history table (**Table ??**) with the Phase 1 predicate row (v1.0.0, June 20 2026, the first-in-human single-arm combined IND/IDE that established the RP2D and device feasibility [1]) and this protocol row (v1.1.0, June 23 2026, the Phase 2 multicenter randomized controlled design with the upgraded Phase 0 gate, single-IRB governance, the co-investment model and capital firewall, and the confirmatory group-sequential framework).*

12.3 Abbreviations

[DRAFTING INSTRUCTION] *Build the two-column-pair abbreviations table (**Table ??**) resolving every abbreviation used across the protocol (AE, BICR, CONSORT, CRF, CRO, ctDNA, CTCAE, DSMB, ECOG, EORTC, FDA, GCP, GJ, HJ, HPB, HR, ICH, IDE, IND, ISGPS, ISM, ITT, KRAS, LLM, MDR, MHRA, mITT, MPR, MTD, NMPA, OS, PACIF, PDAC, PFS, PJ, PMDA, PP, PV, QoL, R0, RCT, RP2D, SAE, SaMD, sIRB, SMA, SMV, TGA, USL, VVUQ).*

13 References and Back Matter

[DRAFTING INSTRUCTION] *Follow the Phase 2 publication section `final-protocol/publication/sections/sec-12-ref` and the Phase 1 model file `trial-protocol/draft-protocol/sections/sec-12-references-backmatter.tex`. Render the References via `ieeetr` from `references.bib`: the five daraxonrasib program entries; the three main documents on which this protocol builds (the 2030 PDAC sixty-second simulation [13], the 21 CFR part 312 Physical AI adaptation [2], and the H.R. 9510 bill [6]); the Phase 1 predicate [1]; the directly relevant author works; the clinical references; the randomized-trial and reporting-method references; and the FDA guidance, the 21 CFR parts (11, 50, 54, 812) and 45 CFR 46, and the IEC / ASME / NASA / IEEE standards with the NIH template. Source the back-matter content from*

trial-protocol/nih-protocol/10_references_and_footnotes.md. Give each main document a brief one-line explanation.

Acknowledgments

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Author and ORCID

[DRAFTING INSTRUCTION] *Kevin Kawchak, CEO ChemicalQDevice, ORCID 0009-0007-5457-8667; render the ORCID iD mark via \orcidicon and the ORCID URL, with the correspondence email.*

Ethical Disclosures

[DRAFTING INSTRUCTION] *No real patients, PHI, or IRB approval; the exemplar PAT-PDAC-0002 is synthetic; all clinical figures derive from the author's simulation sources and the Phase 1 predicate; not an active IND or IDE and not medical or regulatory advice.*

Data Availability

[DRAFTING INSTRUCTION] *GitHub (kevinkawchak/physical-ai-oncology-trials/.../trial-phase-2, repository v4.1.0) plus Zenodo; deterministic seed; no PII; CC BY 4.0; DOI 10.5281/zenodo.xxxxxxx.*

Cite This Article

[DRAFTING INSTRUCTION] *Kawchak K. A Phase 2, Multicenter, Randomized, Controlled Clinical Trial Protocol of On-Premises LLM-Directed Robotic Pancreaticoduodenectomy with Perioperative Daraxonrasib (RMC-6236) in KRAS-Mutated Pancreatic Ductal Adenocarcinoma. Zenodo. 2026; 10.5281/zenodo.xxxxxxx.*

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